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SUPERSEDING
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MILITARY SPECIFICATION

PROJECTILE, 4.2 INCH, HE, M329A2 METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Projectile, 4.2 Inch, HE, M329A2 Metal Parts Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition, General Specification for
MIL-I-6868	- Inspection Process, Magnetic Particle
MIL-M-6867	- Magnetic Inspection Units

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-410	- Qualification of Inspection Personnel, (Magnetic Particle and Penetrant)
MIL-STD-1167	- Ammunition Data Card
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components

FSC 1315

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in approving this document should be addressed to: Commander, Frankford Arsenal, ATTN: SARFA-MDM, Phila., PA 19137 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1 (Cont'd)

DRAWINGS

Armament Command

F11738359	- Metal Parts Assembly - 4.2 Inch, HE, M329A2
F8848553	- Pallet
LI 11738359	- Inspection Equipment List for Projectile, 4.2 Inch, HE, M329A2, Metal Parts Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI-B46.1	- Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI) 1430 Broadway, New York, NY 10018.)

ASTM E 8	- Tension Testing of Metallic Materials
ASTM E 10	- Brinell Hardness of Metallic Materials
ASTM B 117	- Salt Spray (Fog) Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Stress relief.-Each body assembly shall be subjected to a stress relief heat treatment immediately after application of the welded overlay band at a minimum temperature of 700 degrees F for a minimum time of thirty minutes. No subsequent cold working is permitted after final stress relief.

3.3 Mechanical properties.

3.3.1 Body.-Mechanical properties specified on the applicable drawing shall apply.

3.4 Band seat.-The band seat shall be clean and properly prepared for banding.

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3.5 Rotating band application.-The rotating band shall be overlay welded using the gas metal arc welding process in accordance with the applicable drawings and specifications. The gas metal arc process means the use of a consumable electrode with a non-electrode auxilliary wire. The width of the band shall be obtained by oscillating the welding head and the auxilliary wire feed system (see 6.7).

3.5.1 Rotating band seat.-Body material (steel) under the rotating band shall not extend into the groove diameter of the band.

3.6 Metal defects.

3.6.1 Body.-The body, including the cavity, shall be free from cracks, splits, bursts, cold shuts, pipe,porosity, inclusions, folds, seams and other metal defects.

3.6.2 Body cavity.-The body cavity shall be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, pits and sharp edges.

3.6.3 Visual standards.-Visual standards for metal defects of paragraph 3.6.2 will be established for each contract after the first month's production, or whenever requested by the procuring activity (see 6.1).

3.7 Proving ground functional test.-Projectiles, when tested at excess and service pressure shall show no loss or break up of metal parts components in flight.

3.8 Workmanship.-The requirements for workmanship are as shown on the applicable drawings, in the applicable specifications and the following:

3.8.1 Material defects (Obturator).-The obturator shall be free from flash and porosity.

3.8.2 Surface finish.-The requirements for surface finish are as detailed on the applicable drawing.

3.8.3 Threads.-Threads shall be full and undamaged for the entire minimum length or depth as specified on the applicable drawing.

3.8.4 Painting.-Painting shall comply with the requirements of the applicable drawing. All paint shall be dry to the touch before testing and shall be in accordance with the applicable specification.

3.8.5 Base cover braze or welding.-The braze or welding shall be sound, and the brazing or welding operation shall be so controlled that there will be adequate bonding between the projectile base and base cover.

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3.8.6 Burr.-No part shall have a burr which might interfere with the assembly of the round or which might be injurous to personnel handling the item.

3.8.7 Foreign matter.-No part or assembly shall contain dirt, grease, oil, rust, chips, corrosion or other foreign matter during inspection or testing. Particular attention shall be given to the charge cavity of the assembly, to insure that no grit, scale or other foreign matter remains therein after final cleaning.

3.8.8 Protection of rotating band.-The machined rotating band shall be protected from being nicked, burred or otherwise damaged in subsequent handling prior to shipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550, except as otherwise specified herein.

4.1.3 Inspection criteria.-

a. Successful passage of all inspection required by the drawings and specifications shall be considered as a condition precedent to performing the Proving Ground functional tests (see 3.7).

b. Acceptance or rejection of a lot by the Government shall be contingent on successful passage of the Proving Ground functional tests.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 55 metal parts assemblies, as follows:

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4.2.1 Cont'd)

45 metal parts assemblies complete including paint
5 metal parts assemblies complete except paint
5 sets component parts unassembled and unpainted

Forty of the above complete and painted metal parts assemblies shall be forwarded to a designated proving ground for ballistic testing. The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials including packaging and packing shall be the same as used in regular production and shall be obtained from the same source of supply. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever a change occurs in manufacturing process, procedure, material used, drawings, specifications or source of supply such as to significantly affect product uniformity as determined by the Government.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample will be inspected for all the requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications will result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Stress-relieved lots.-Stress-relieved lots shall contain cold-shaped bodies from not more than one heat of steel, (after application of the welded overlay band), that has been cold shaped by the same process and stress relieved in the same furnace at the same termperature for the same length of time. The stress relieved lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.3 Projectile lots.-When it has been determined that bodies of a heat treated or stress relieved lot have complied with the mechanical properties requirement of the applicable drawing, they may be grouped for further processing.

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4.3.1.4 Shipping lots.-Shipping lots shall consist of projectile assemblies produced by one supplier, under one contract, in one unchanged process, in accordance with the same drawings, revisions, under conditions of a continuous production, free of interruptions other than those due to end of shift day or work week. Shipping lots shall be assigned a lot number in accordance with MIL-STD-1168. The shipping lot thus constituted shall be the basis for selection of samples for the proving ground functional test. Lot size shall be determined by the procuring contracting officer.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects, examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. Continuous sampling plans in accordance with standard MIL-STD-1235 may be used if approved by the procuring activity. An Acceptable Quality Level (AQL) of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual inspection. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis for surface roughness determination.

4.3.2.1 Tool control dimensions.-

a. Dimensions marked tool control shall be gaged at the beginning or end of each working day (24 hrs), during First Article Inspection and for the first component after each tooling change, adjustment or repair. Where destruction of components is necessary to gage such dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production.

b. When the contractor has satisfied (a) above, visual inspection of tool control dimensions will be permitted. Visual inspection shall be performed every two (2) hours for each machine in operation. If any tool control dimension(s) in in doubt during visual inspection, the dimension(s) shall be immediately gaged (see "a" above) to determine acceptability. Corrective action will immediately be instituted by the contractor whenever a defective tool control dimension is found.

4.3.2.2 Body, (before banding) Drawing 11738399, covering a detail of Drawing F11738360.

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual (Probe Light)

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4.3.2.2 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Runout of rotating band seat with bourrelet and rear of band	Gage (see 4.4.2)
102. Diameter of rotating band seat	Gage
103. Location of rotating band seat	Gage (see 4.4.2)
104. Width of rotating band seat, min.	Gage
105. Foreign matter on rotating band seat	Visual
MINOR	
201. Surface finish improper	Visual (see 6.2)
202. Radii or chamfers missing or incorrect	Visual (see 6.2)
203. Burr	Visual
204. Foreign matter (except as otherwise classified)	Visual

4.3.2.3 Body Assembly, Drawing F11738360, covering a detail of Drawing F11738359.

CRITICAL	
1. Metal Defective	Visual (Probe Light)
MAJOR	
101. Runout of pitch diameter of fuze thread with bourrelet, rotating band and front face	Gage
102. Runout of pitch diameter of boss thread with bourrelet, rotating band and rear face	Gage
103. Runout of ogive with bourrelet and rotating band	Gage
104. Runout of body undercut with bourrelet and rotating band	Gage
105. Runout of boattail with bourrelet and rotating band	Gage
106. Runout of obturator band seat with bourrelet and rotating band	Gage
107. Diameter of rotating band teeth	Gage
108. Diameter forward end of rotating band	Gage
109. Diameter rear end of rotating band	Gage
110. Diameter between rotating band teeth	Gage
111. Diameter of body undercut	Gage
112. Diameter of bourrelet	Gage
113. Diameter of obturator band seat	Gage

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4.3.2.3 (Cont'd)

	Defect	Method of Inspection
114.	Diameter of base cover	Gage
115.	Length from front face to forward basic diameter of ogive	Gage
116.	Length from front face to rear basic diameter of ogive	Gage
117.	Length from front face to bourrelet	Gage
118.	Length from front face to front of rotating band teeth	Gage
119.	Length from front face to front of obturating band seat	Gage
120.	Length from front face to rear face	Gage
121.	Length of rotating band teeth	Gage
122.	Length from rear face to basic diameter on boattail	Gage
123.	Length of thread boss	Gage
124.	Length from front of rotating band to front of rotating band teeth	Gage
125.	Length of angle rear of rotating band	Gage
126.	Angle rear of obturator band seat	Gage
127.	Variation in wall thickness	Gage
128.	Wall thickness, min.	Gage
129.	Width of rotating band teeth	Gage
130.	Width of obturator band seat	Gage
131.	Thickness through base	Gage

MINOR

201.	Diameter of thread boss undercut	Gage
202.	Width of thread boss undercut	Gage
203.	Width of undercut rear of rotating band teeth (if applicable)	Gage
204.	Depth of undercut rear of rotating band teeth (if applicable)	Gage
205.	Surface finish improper	Visual (see 6.2)
206.	Radii or chamfers missing or incorrect	Visual (see 6.2)
207.	Staking notches missing or incorrect	Visual (see 6.2)
208.	Burr	Visual
209.	Foreign matter	Visual

4.3.2.4 Metal Parts Assembly, Drawing F11738359 (after paint).

CRITICAL

1.	Metal defective	Visual (Probe Light)

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4.3.2.4 (Cont'd)

Defect	Method of Inspection
2. Magnetic particle stamp missing (signifying failure to pass magnetic particle inspection)	Visual
3. Pool of paint in cavity	Visual (Probe Light)
MAJOR	
101. Weight	Approved Scale
102. Pitch diameter of fuze thread, max.	Gage
103. Pitch diameter of fuze thread, min.	Gage
104. Minor diameter of fuze thread, min.	Gage
105. Pitch diameter of boss thread, min.	Gage
106. Pitch diameter of boss thread, max.	Gage
107. Major diameter of boss thread, max.	Gage
108. Length of fuze thread, min.	Gage
109. Length of fuze thread, max.	Gage
110. Rotating band damaged	Visual
111. Foreign matter in cavity	Visual
MINOR	
201. Paint on fuze thread	Visual
202. Paint on rotating band bearing surfaces	Visual
203. Protective coating on interior surfaces inadequate or damaged	Visual
204. Protective coating on exterior surfaces inadequate or damaged	Visual
205. Corrosive preventive compound missing from threads	Visual
206. Protector cap missing	Visual
207. Radii or chamfers missing or incorrect	Visual
208. Marking missing, incorrect or illegible	Visual
209. Burr	Visual
210. Foreign matter (except as otherwise classified)	Visual

4.3.2.5 Obturator, Drawing C11738362.

CRITICAL	
None defined	
MAJOR	
101. Presence, location and depth of slits	Visual (see 6.9)
102. Material defective	Visual

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4.3.2.5 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Excessive flash	Visual
202. Porosity	Visual

4.3.2.6 Pallet, Drawing F8848553.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Pallet damaged (unserviceable)	Visual
202. Sleeve missing	Visual
203. Improperly packed	Visual
204. Strapping missing, incorrect or broken	Visual/Manual
205. Marking missing, incorrect or illegible	Visual

4.3.2.7 Profile alignment gage.-Each metal parts assembly after paint shall be gaged in a Government certified chamber gage. Failure to comply with the requirements of the applicable drawing shall be cause for rejection of the metal parts assembly.

4.3.3 Inspection testing.

4.3.3.1 Mechanical properties. (see 6.4)

4.3.3.1.1 Hardness.

4.3.3.1.1.1 Stress relieved lots.-Twenty-five (25) cold-shaped bodies from the first batch of each stress relieved lot (as defined in 4.3.1.2) processed through the stress relief furnace after application of the welded overlay band and 5 bodies from each hour's production thereafter for the balance of the stress relieved lot, after application of the welded overlay band, shall be subjected to a hardness test to assess uniformity of the process.

4.3.3.1.2 Tension test.

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4.3.3.1.2.1 Stress-relieved lots.-Two bodies, one of the hardest and one of the softest, as determined by the hardness test shall be subjected to this test. Failure of any tension test speciment to comply with the mechanical properties requirement of the applicable drawing shall be cause for rejection of the lot. When it has been determined that the mechanical properties comply and furnace temperature and time have been established, the furnace temperature and time shall be maintained throughout the stress-relieved lot. The furnace temperature may vary within twenty-five degrees Fahrenheit, but not below the minimum temperature of 700 degrees F, and the soak time of 30 minutes minimum. Furnace temperature and time will be recorded every hour to assess adequate control of stress relief equipment. The furnace temperature will be calibrated every eight hours by an independent method.

4.3.3.2 Magnetic particle inspection.

4.3.3.2.1 Band seat.-The first five banded bodies, after the welded overlay band has been applied and removed, for each welding machine in operation shall be subjected to this test, to determine compliance with the visual standards set by the responsible technical engineering agency. Reduced sampling of one in twenty banded bodies may be implemented after five consecutive banded bodies per each welding machine in operation have been visually examined and found acceptable. Reduced sampling of one in two-hundred banded bodies may be implemented after five consecutive samples of one in twenty banded bodies per welding machine in operation have been visually examined and found acceptable. Reduced sampling of one banded body per each welding machine in operation per lot may be implemented after five consecutive samples of one in two hundred banded bodies per each welding machine in operation have been visually examined and found acceptable. Should two consecutive failures occur in any welding machine(s), in any reduced sample size, the machine(s) shall revert back to the original first five consecutive banded bodies per that welding machine(s) and the sequence of reduced sampling will also apply.

4.3.3.2.2 Body assembly.-Each finished machined body assembly shall be subjected to a magnetic particle inspection. Failure of any body assembly to comply with the requirements of 3.6.1, except as determined by 4.3.3.2.1 shall be cause for rejection. Body assemblies which comply shall be stamped with the letter "M" as indicated on the drawing.

4.3.3.3 Base, cover braze or weld.-Two metal parts assemblies from each one hours production or fraction thereof per brazing or welding machine in operation shall be subjected to this test. Failure of one or more of the brazed or welded base covers to meet the requirements of 3.8.5 shall be cause for rejection of the production represented by the sample.

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4.3.3.4 Band chemical analysis.-One body after banding from each 8 hours production or fraction thereof for each welding machine in operation shall be subjected to this test. Failure to comply with the requirements of the drawing shall be cause for rejection of the production represented by the sample.

4.3.3.5 Cleaning and application of protective coating.-The sample size and acceptance criteria for the inspection of cleaning the item and application of protective coating shall be as specified in the applicable specification. Failure to comply with the requirements shall be cause for taking the corrective action recommended therein.

4.3.3.5.1 Salt spray.-The sample size and acceptance criteria for salt spray testing shall be in accordance with the applicable specification. Failure of the sample specimens to comply with the requirements of the applicable drawing shall be cause for taking the corrective action recommended by the applicable specification.

4.3.4 Proving ground functional test.-The following proving ground tests will be performed by the Government.

4.3.4.1 First article sample.

4.3.4.1.1 Initial production sample.-The quantities listed in Table I will be ballistically tested in the phases indicated.

TABLE I

Phase	Quantity
Excess pressure	25
Service pressure	15

4.3.4.2 Production lot sample.-The quantities listed in Table II shall be ballistically tested in the phases indicated.

TABLE II

Phase	Quantity
Excess pressure	10
*Service pressure	10

*After three(3) consecutive lots have been ballistically tested and accepted, ballistic testing for this phase may be discontinued with the approval of the procuring activity. However, implementation of ballistic testing will be instituted whenever directed by the procuring contracting officer.

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4.3.4.3 Excess pressure.-Samples provided in accordance with Tables I and II will be subjected to this test. Failure of any metal parts assembly to comply with 3.7 shall be cause for rejection of the initial production sample of the production lot represented by the sample.

4.3.4.4 Service pressure.-Samples provided in accordance with Tables I and II will be subjected to this test. Failure of any metal parts assembly to comply with 3.7 shall be cause for rejection of the initial production sample or the production lot represented by the sample.

4.3.5 Inspection equipment.-Index of Inspection Equipment Lists LI-11738359 identify the inspection equipment required to perform the examinations and tests prescribed herein. Equipment design shall be in accordance with code designation on the applicable LI (Reference is made to Inspection Equipment List for definition of coding format).

4.4 Test methods and procedures.

4.4.1 Mechanical properties. (see 6.4)

4.4.1.1 Hardness.

4.4.1.1.1 Body.-After application of the welded overlay band and after final stress relief and before final machining, the hardness test shall be performed on the body approximately one inch forward of the rotating band. This test shall be made using the procedure and equipment for performing this test as prescribed by ASTM Method E-10.

4.4.1.2 Tension test.

4.4.1.2.1 Yield strength and elongation.-From the samples selected in accordance with 4.3.3.1.2.1, seven tension test specimens conforming to the largest obtainable standard "R" type specimen as prescribed by ASTM Method E-8 shall be taken from each sample as follows: two longitudinal tension test specimens from the ogive area 180 degrees apart; two longitudinal tension test specimens in the body area between the ogive and zone 1, 180 degrees apart; two longitudinal tension test specimens midwall of the band seat area, with the gage length centered under the band, 180 degrees apart; and one transverse tension test specimen from the rear face of the body approximately three quarters of an inch from the centerline. The number of specimens and their location is to determine the acceptability of the mechanical properties requirements of the applicable drawing and with the approval of the procuring agency, the Government shall not be restricted from testing additional specimens at other locations to determine that the mechanical properties of the projectile conform to the drawing requirements. The criteria for performing this test shall be as prescribed by ASTM Method E-8 except that elongation shall be measured to the nearest thousandth (0.000) prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth (0.0) percent.

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4.4.2 Rotating band seat.-The contractor may use any control method that will assure compliance with 3.5.1 and the dimensional requirements of the applicable drawing. The inspection plan must include all of the required inspection criteria to provide assurance that the dimensional aspects of the applicable drawing will be equal to that provided by the inspection procedures associated with the applicable characteristics listed in 4.3.2.2. When the contractor established a control method to assure his meeting these requirements, the manufacturing procedure and inspection plan shall be forwarded to the responsible tenchical engineering agency for approval prior to its implementation into production.

4.4.3 Magnetic particle inspection.

4.4.3.1 Band seat.-The sample provided in accordance with 4.3.3.2.1 after heat treatment shall have the band removed by machining the band and chemically etching without damaging the base metal of the band seat (see 6.3). The procedures for performing this test shall be as specified in MIL-I-6868 for wet flourescent method by either the continuous or residual methods. The band seat shall be inspected after inducing a circular magnetic field in the body and after inducing a longitudinal magnetic field in the body.

4.4.3.2 Body.-The procedures for performing this test shall be as specified in MIL-I-6868, for wet flourescent method by either the continuous or residual methods. The entire body, inside and outside, shall be inspected after inducing a circular magnetic field in the body and after inducing a longitudinal magnetic field in the body.

4.4.3.3 Suspension.-The solid content of the suspension shall be maintained at the level recommended by the suspensoid manufacturer. The suspension shall be checked at the start of each day's production and at intervals of not over 4 hours of continuous operation, utilizing the procedures and equipment prescribed by Section 4 of MIL-I-6868. When performing this test for solid content of the suspension, if it is determined that it does not comply, all bodies that have been tested and accepted since the last satisfactory test was performed shall be retested after the suspension has been brought to the proper level. The magnetic particle test shall be performed on equipment that complies with the requirements of specification MIL-M-6867. The operator(s) of the equipment and the inspector(s) making the determinations shall be certified in accordance with MIL-STD-410.

4.4.4 Band chemical analysis.-From the bodies selected in 4.3.3.4 sample chips shall be obtained from each band by machining prior to finish diameter. All chips shall be collected and identified with the shell from which the chips were machined. A minimum 5 gram sample from each complete specimen thus obtained from each band shall be dissolved and the analysis made from the individual solutions. The sample may not be cleaned with a magnet.

4.4.5 Protective coating.-The procedures for testing the cleaning and surface treatment methods for components and assemblies shall be in accordance with the methods and criteria for testing as specified on the applicable specification.

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4.4.5.1 Salt spray.-The salt spray test prescribed by the applicable drawing shall be performed utilizing the equipment and procedures specified by ASTM Method B-117.

4.4.6 Base cover braze or weld.-The test of the base cover braze or weld shall be performed as follows: Visually inspect the base cover weld and select an area that appears to have the weakest braze or weld;insert a 5/8" chisel in this a rea and apply force by use of a 1 pound steel hammer and an approximately 1 foot stroke. If the braze or weld breaks loose after two strokes, the body assembly will be classed defective. If the braze is found to be secure, the burr made by the chisel may be knocked back into position and accepted.

4.4.7 Proving ground functional test.

4.4.7.1 Excess pressure.-The ballistic test projectiles inert loaded and dummy fuzed will be fired in an acceptable standard gun based upon current gun tube condemnation criteria. A propellant charge adjusted to produce 109 to 117 percent of the upper pressure limit of the propellant will be used. Photographic equipment (smear camera) will be set up near the gun muzzle to record the flight of t he projectile. Visual observation will be made to assure impacts in target area. Velocity and chamber pressure will be recorded.

4.4.7.2 Service pressure.-The ballistic test projectiles inert loaded and dummy fuzed will be fired alternately with projectiles from a reference lot for comparison. The projectiles will be fired in an acceptable standard gun based upon current gun tube condemnation criteria. The projectile shall be fired for accuracy at a range selected by the proof director. Photographic equipment (smear camera) will be set up near the gun muzzle to record the flight of the projectile. Visual observation will be made to assure projectile impacts in the target area. Velocity and chamber pressure will be recorded.

5. PREPARATION FOR DELIVERY

5.1 Level C.-Minimum military pack.

5.1.1 Level C.-Projectile metal parts assemblies shall be palletized in accordance with Drawing F8848553.

5.2 Packaging, packing and marking.-Packaging, packing and marking shall be in accordance with MIL-STD-1169.

5.3 Data cards.-Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Visual standards.-Visual standards to be established at contractor's plant in accordance with 3.6.3.

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6.2 Visual examination qualification.-When compliance with the applicable requirements are in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.3 Suggested etching procedures for removing aluminum bronze overlay

A. The following solution -

500 grams Chromic Oxide (Commercial flake)
50 ml Sulphuric Acid (Technical grade)
950 ml Water

can be used by passing current through the solution (50 amperes dc) with the immersed shell as the anode and the container as the cathode. Approximately 0.005 to 0.010 inch thick overlay material will require approximately two to three hours in fresh solution.

B. The solution can also be used without an external current source by immersing the shell for approximately 8 to 12 hours. An active solution would clean all pits.

6.4 Alternative mechanical properties testing.-Alternative procedures for mechanical properties testing may be used by the contractor when such testing and procedures provide as a minimum the quality assurance specified herein. Prior to applying the alternative procedures, the contractor shall describe them in a written proposal and shall demonstrate for the approval of the responsible technical agency that their effectiveness is equal to or better than the quality assurance provision specified herein.

6.5 Experience to date has indicated that 1340 steel requires close control over the welding process and stress relief to prevent cracks after banding.

6.6 Caution should be exercised in the final machining operation due to the probability of encountering ovality in the body after welding the overlay band.

6.7 Recommend welding operation be guided by Frankford Arsenal "Process Recommendations for Banding the 4.2 Inch, H.E., M329 Projectile".

6.8 Ordering data.-Procurement documents should specify the title, number and the date of this specification.

6.9 When inspecting for slits, the obtruator shall be distorted near slits in such a manner so that the slits are visible. Should the depth of the slits be in doubt as a result of visual examination, it may be measured or gaged to determine acceptability.

Custodian:

Army - MU

Preparing activity:

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Project No. 1315-A283

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